L01-R01 — one continuous density proof

Purpose

Prove the conditional density used in the lecture from the supplied joint Gaussian density.

Endpoint

$$p(x_1|x_2) = \mathcal{N}_{d_1}(x_1; \mu_1 + B(x_2 - \mu_2), S)$$

, identified by integrating and dividing a proved normalized joint-density product.

Entry / assumed context

Block Gaussian notation, conditional mean/covariance and centered residual from the lecture; matrix algebra, strict positive definiteness, determinants, density normalization. The lecture's residual-independence proof is not a premise of this argument.

Narrative arc

Orientation → identify the exponential and its coefficient → choose and justify the residual coordinates → eliminate the two covariance cross blocks → prove positivity and the inverse identity → calculate the two quadratic terms → split the exponential, determinant and dimension factors → integrate and divide → interpret observation dependence.

Visual and notation anchors

$$H = \Sigma_{22}, \quad BH = \Sigma_{12}$$
$$u = x_2 - \mu_2, \quad r = x_1 - \mu_1 - Bu$$
$$Q = (x-\mu)^T \Sigma^{-1} (x-\mu)$$

The map T subtracts the predicted deviation. $D = T\Sigma T^T$ is block diagonal in the residual/observed coordinates. C is the coefficient before the density's exponential.

Summary design brief

No summary slide: explicit instructor recording convention. P17 interprets the established mean and covariance as its own teaching move.

Continuity / entry context

Inspected rendered L01 handout p.48: conditional formula beside a labeled mean proof; the zero-covariance step retains its joint-Gaussian warrant. Inspected squared-loss proof source: carried-forward expansion, local operation "Take conditional expectation", separately labeled cross-term and residual energy, interpretation in a lower key-point slot. Here, preserve operations and warrants beside equations and give useful meaning in every reveal state.

Constraints / open pressure

All 18 units require production rebuild. Object kinds describe semantics: they do not require bordered production cards around every proof step. Prefer plain mathematical regions with nearby operation labels; reserve borders for a genuine orientation/comparison or key-point role. In production use math-rows only for standalone equations, never prose containing inline mathematics; prose must keep its inline symbols within the sentence. Pass raw interpretation content to content-block key-points, which already supplies its wrapper. P0 preserves the instructor's L01-R01 number and exact subtitle. No notebook or live-proof relocation. Keep full SPD, inverse, quadratic, determinant/dimension and integral/ratio arguments. No unexplained "normalizer", "proof obligation", or "Prepared live". Formula rows require generous vertical separation, including recall cards; do not pack them through markup line breaks or shrink them to preserve a count. Timing remains unmeasured. Exact source record: math_derivation.md, Murphy Book 2 (2023), online December 10, 2025, §2.3.1.5, pp.25–27; explicit elimination/SPD/normalization completion is course construction.

Quadratic separation of a Gaussian

Teaching move

Identify the recording and its concrete purpose.

Layout sketch

subtitle

Content objects

subtitle card

A proof of the formula for the conditional density of a Gaussian

Reveal sketch

Static: subtitle

Design constraints

Production title metadata number is L01-R01. Exact subtitle as written. No section title follows this slide.

From the lecture to this proof

Teaching move

Recall the supplied model on the left, state the density result to justify on the right, and orient the proof.

Layout sketch

lecture	**goal**
kp	

Content objects

lecture — From the lecture recall-card

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}_d(\mu, \Sigma)$$

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$\Sigma = \Sigma^T \succ 0, \quad d = d_1 + d_2$$

From: L01: Gaussian blocks, conditioning and centered residual

goal — To prove here card

$$p(x_1|x_2) = \mathcal{N}_{d_1}(x_1; m(x_2), S)$$

The mean $m(x_2)$ and covariance S obtained in the lecture.
Derive this complete density from $p(x_1, x_2)$.

kp — Key points key-points

Fix the observed value x_2; the density describes the missing block X_1.
We will account for the exponential and its coefficient, then identify the conditional by integration and division.

Reveal sketch

Static: lecture, goal, kp

Design constraints

Two orientation columns, then a distinct key-point region visible from the start. Each formula is a separate generously spaced row, especially in the supplied-model recall. X_i, μ_i have dimension $d_i \geq 1$; covariance blocks match. Recalled conditional is a destination, never a proof premise.

The density contains an exponential and a coefficient

Teaching move

Identify the exact quadratic and coefficient to transform, and explain what their transformations must achieve.

Layout sketch

DESIGN LAYOUT

Joint-density equation above two adjacent targets: quadratic on the left, coefficient on the right. Distinct lower key-point slot.

Content objects

density — Joint Gaussian density — card

$$p(x_1, x_2) = C \exp\left(-\frac{1}{2}Q\right)$$

quadratic — The quadratic in the exponent — card

$$Q = (x-\mu)^T \Sigma^{-1} (x-\mu)$$

Target: residual contribution + observed-deviation contribution.
Residual: $x_1 - m(x_2)$; observed deviation: $x_2 - \mu_2$.

coefficient — The factor before the exponential — card

$$C = (2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}}$$

Target: coefficient for the residual density × coefficient for the observed-block density.

kp — Key points — key-points

Writing the exponential as a product will not establish the full density product until its coefficient also matches.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	density, quadratic, kp	—	Initial meaningful context
STATE 2	+ coefficient	entry emphasis	Identify the second part of the same density that must match.

Design constraints

“Residual” means deviation of x_1 from its observation-dependent conditional mean; “observed deviation” means $x_2 - \mu_2$. These labels precede the coordinate definitions on P4. Do not call either a “normalizer” or an “obligation”.

Choose B to remove the cross covariance

Teaching move

Give the inverse a purpose: solve the cross-block cancellation equation, proving that the observed-block covariance permits its solution.

Layout sketch

DESIGN LAYOUT

Cancellation target first, then adjacent proof regions: positivity test on the left, solved coefficient on the right. The two regions share a baseline and sufficient space for the full test-vector calculation. Plain equations and local operation labels, without a separate bordered card for every substep. Meaning accompanies the opening; result accompanies the solution in the distinct key-point slot.

Content objects

target — Subtract B times the observed deviation: cancel the cross block — card

$$H := \Sigma_{22}$$

$$\Sigma_{12} - BH = 0 \quad \Leftrightarrow \quad BH = \Sigma_{12}$$

positive — Joint strict positivity applies to observed-block vectors — card

$$z \in \mathbb{R}^{d_2} \setminus \{0\}$$

$$z^T H z = \begin{pmatrix} 0 \\ z \end{pmatrix}^T \Sigma \begin{pmatrix} 0 \\ z \end{pmatrix} > 0$$

$$H = H^T \succ 0 \quad \Rightarrow \quad H^{-1} \text{ exists}$$

solution — Solve the cancellation equation; transpose it — card

$$B = \Sigma_{12} H^{-1}, \quad H B^T = \Sigma_{21}$$

meaning — Key points — key-points

Cancel the covariance cross blocks so its inverse acts separately on residual and observed coordinates. This will give the two terms in Q.

result — Key points — key-points

Strict positivity makes this choice well defined. It is the coefficient already used for the conditional mean in the lecture.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	target, meaning	—	Initial meaningful context
STATE 2	+ positive	entry emphasis	Justify the inverse needed to solve this equation.
STATE 3	+ solution, result	entry emphasis	Right-multiply by the justified inverse; use symmetry of H inverse and Sigma21=Sigma12 transpose.

Design constraints

The cancellation target is displayed before the positivity calculation. B has size $d_1 \times d_2$. Do not introduce S or its inverse here.

Subtracting the predicted deviation produces the residual

Teaching move

Make the coordinate change intelligible before using it to transform the covariance.

Layout sketch

coordinates

map

target

kp

Content objects

coordinates — Observed deviation and centered residual card

$$u := x_2 - \mu_2$$
$$m(x_2) := \mu_1 + Bu, \quad r := x_1 - m(x_2)$$

map — Apply the subtraction while retaining the observed deviation card

$$T := \begin{pmatrix} I_{d_1} & -B \\ 0 & I_{d_2} \end{pmatrix}$$
$$T(x-\mu) = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} x_1 - \mu_1 \\ u \end{pmatrix} = \begin{pmatrix} r \\ u \end{pmatrix}$$

target — To show: the residual and observed contributions card

$$Q = r^T S^{-1} r + u^T H^{-1} u$$

kp — Key points key-points

The first new coordinate is error around the predicted mean; the second is the observed deviation. These are the two contributions sought in the exponent.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	coordinates, kp	—	Initial meaningful context
STATE 2	+ map, target	entry emphasis	Represent the meaningful subtraction as a block matrix operation and write the exact quadratic target.

Design constraints

Use the names residual and observed deviation adjacent to their definitions. Identity and zero block dimensions follow d1,d2 and may subsequently omit subscripts. No independence claim. S in the target denotes the residual covariance recalled on P1; its formula is obtained on P6 and its positivity is proved on P8. The target is not a premise.

The coordinate change is reversible

Teaching move

Verify the inverse of this specific coordinate change and explain why retaining the observation loses no information.

Layout sketch

recall

undo

check

kp

Content objects

recall — Carried forward recall-card

$$T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}, \quad T(x-\mu) = \begin{pmatrix} r \\ u \end{pmatrix}$$

From: P4: residual coordinates

undo — Add the predicted deviation back card

$$U := \begin{pmatrix} I & B \\ 0 & I \end{pmatrix}, \quad U\begin{pmatrix} r \\ u \end{pmatrix} = \begin{pmatrix} r + Bu \\ u \end{pmatrix} = x - \mu$$

check — Check both matrix products card

$$TU = \begin{pmatrix} I & B - B \\ 0 & I \end{pmatrix} = I$$

$$UT = \begin{pmatrix} I & -B + B \\ 0 & I \end{pmatrix} = I \quad \Rightarrow \quad T^{-1} = U$$

kp — Key points key-points

The residual together with the observation determines the original centered vector. This invertibility will preserve strict positivity and let us recover the inverse covariance.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall, undo, kp	—	Initial meaningful context
STATE 2	+ check	entry emphasis	Verify that adding back the predicted deviation is the two-sided inverse.

Subtract the observed rows to cancel the upper-right block

Teaching move

Work the first elimination and identify the remaining first-block covariance expression.

Layout sketch

recall

multiply

result

kp

Content objects

recall — Carried forward recall-card

$$\Sigma = \begin{pmatrix} \Sigma_{11} & BH \\ HB^T & H \end{pmatrix}, \quad T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}$$

From: P3: B H=Sigma12; P4: T

multiply — First block row minus B times the second card

$$T\Sigma = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} \Sigma_{11} & BH \\ HB^T & H \end{pmatrix}$$
$$= \begin{pmatrix} \Sigma_{11} - BHB^T & BH - BH \\ HB^T & H \end{pmatrix}$$

result — Name the remaining first block card

$$S := \Sigma_{11} - BHB^T = \Sigma_{11} - \Sigma_{12}H^{-1}\Sigma_{21}$$
$$T\Sigma = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix}$$

kp — Key points key-points

The chosen B cancels the upper-right block exactly. The lower-left block remains because only the row operation has been applied.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall, kp	—	Initial meaningful context
STATE 2	+ multiply	entry emphasis	Work the targeted cross-block cancellation without hiding the other blocks.
STATE 3	+ result	entry emphasis	Recognize the Schur-complement expression obtained in the lecture.

Design constraints

S is defined here by the actual product; its strict positivity is not yet established. Keep all four product blocks visible long enough to check.

The matching column operation cancels the lower-left block

Teaching move

Finish the two-sided covariance transformation and exhibit the separate residual/observed covariance blocks.

Layout sketch

recall

multiply

diagonal

kp

Content objects

recall — Carried forward recall-card

$$T\Sigma = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix}, \quad T^T = \begin{pmatrix} I & 0 \\ -B^T & I \end{pmatrix}$$

From: P6 and transpose of T

multiply — Apply the same coordinate change to covariance columns card

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix}\begin{pmatrix} I & 0 \\ -B^T & I \end{pmatrix}$$
$$= \begin{pmatrix} S & 0 \\ HB^T - HB^T & H \end{pmatrix}$$

diagonal — Residual block S; observed block H card

$$D := T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}$$

kp — Key points key-points

The row and column operations together give the covariance in the residual/observation coordinates. Its off-diagonal blocks are zero; its inverse will produce the two quadratic terms.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall, kp	—	Initial meaningful context
STATE 2	+ multiply	entry emphasis	Show exactly where the remaining cross block cancels.
STATE 3	+ diagonal	entry emphasis	Name the block diagonal result used in the exponent and determinant calculations.

Design constraints

This is a congruence TΣT transpose. No similarity-transformation or independence detour; the proof uses the displayed matrix identity.

The remaining residual block is positive definite

Teaching move

Justify S as a valid nonsingular covariance and justify all subsequent inverse and square-root operations.

Layout sketch

recall

positiveD

positiveS

kp

Content objects

recall — Carried forward recall-card

$$D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}, \quad \Sigma \succ 0, \quad T \text{ invertible}$$

From: P5 and P7

positiveD — Invertibility keeps a nonzero test vector nonzero card

$$w \neq 0 \quad \Rightarrow \quad T^T w \neq 0$$

$$w^T D w = (T^T w)^T \Sigma (T^T w) > 0$$

positiveS — Restrict the test to the residual block card

$$a \in \mathbb{R}^{d_1} \setminus \{0\}, \quad w = \begin{pmatrix} a \\ 0 \end{pmatrix}$$

$$a^T S a = w^T D w > 0 \quad \Rightarrow \quad S = S^T \succ 0$$

kp — Key points key-points

The supplied joint strict positivity survives the reversible coordinate change. Both S and H therefore have inverses and positive determinants, as the Gaussian density requires.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall, kp	—	Initial meaningful context
STATE 2	+ positiveD	entry emphasis	Apply joint strict positivity to the transformed test vector.
STATE 3	+ positiveS	entry emphasis	A vector supported on the first block isolates S.

Invert the covariance identity to reach the exponent

Teaching move

Connect the transformed covariance to the inverse covariance actually appearing in Q, with exact operation order.

Layout sketch

recall

invert

isolate

kp

Content objects

recall — Carried forward recall-card

$$Q = (x-\mu)^T \Sigma^{-1} (x-\mu)$$

$$D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}$$

From: P2: exponent; P7–P8: valid block covariance

invert — Invert the product: reverse its factors card

$$D^{-1} = T^{-T} \Sigma^{-1} T^{-1}$$

$$T^{-T} := \left(T^{-1}\right)^T$$

isolate — Multiply on the left by T transpose and on the right by T card

$$T^T D^{-1} T = \Sigma^{-1}$$

$$\Sigma^{-1} = T^T \begin{pmatrix} S^{-1} & 0 \\ 0 & H^{-1} \end{pmatrix} T$$

kp — Key points key-points

We need the inverse covariance in the exponent. This identity places T next to each centered vector, so the residual coordinates can be substituted directly.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall, kp	—	Initial meaningful context
STATE 2	+ invert	entry emphasis	Use the already justified inverses; do not commute factors.
STATE 3	+ isolate	entry emphasis	Cancel adjacent inverse pairs to obtain the expression needed in Q.

Design constraints

Keep operation labels next to their equalities. This is algebraic inverse notation; no numerical inverse algorithm is introduced.

The joint quadratic becomes a residual term plus an observed term

Teaching move

Perform the promised quadratic split with explicit substitutions and labeled final terms.

Layout sketch

DESIGN LAYOUT

Subordinate recall strip keeps both anchors together. Below it, one wide plain proof region contains the whole inverse-substitution equality, the diagonal block multiplication, and the labeled final split as generously separated rows. Keep local operation labels adjacent, with no per-row bordered wrappers. Interpretation is a separate lower region.

Content objects

recall — Carried forward — recall-card

$$\Sigma^{-1} = T^T D^{-1} T, \quad T(x-\mu) = \begin{pmatrix} r \\ u \end{pmatrix}$$

From: P4 and P9

substitute — Substitute the inverse; then the residual coordinates — card

$$Q = (x-\mu)^T \Sigma^{-1} (x-\mu) = [T(x-\mu)]^T D^{-1} [T(x-\mu)]$$
$$= \begin{pmatrix} r \\ u \end{pmatrix}^T \begin{pmatrix} S^{-1} & 0 \\ 0 & H^{-1} \end{pmatrix} \begin{pmatrix} r \\ u \end{pmatrix}$$

split — Multiply the diagonal blocks — card

$$Q = \underbrace{r^T S^{-1} r}_{\text{residual term}} + \underbrace{u^T H^{-1} u}_{\text{observed term}}$$

kp — Key points — key-points

At fixed x_2, u is fixed and $r = x_1 - m(x_2)$ varies with x_1. The exponent now has precisely a residual contribution and an observed-deviation contribution.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall, kp	—	Initial meaningful context
STATE 2	+ substitute	entry emphasis	Each T acts on the centered vector beside it; retain the whole equality.
STATE 3	+ split	entry emphasis	The zero off-diagonal inverse blocks remove mixed terms.

Design constraints

Keep the first equality whole and use labels adjacent to the two final terms. Do not yet identify a normalized conditional from the exponent alone.

A sum in the exponent becomes a product

Teaching move

Show exactly how the quadratic split transforms the joint density and identify the remaining coefficient calculation.

Layout sketch

recall

product

coefficient

kp

Content objects

recall — Carried forward recall-card

$$Q = r^T S^{-1} r + u^T H^{-1} u$$

From: P10

product — Use exp(a+b)=exp(a)exp(b) card

$$\exp\left(-\frac{1}{2}Q\right) = \exp\left(-\frac{1}{2}r^T S^{-1} r\right) \exp\left(-\frac{1}{2}u^T H^{-1} u\right)$$

$$p(x_1, x_2) = C \exp\left(-\frac{1}{2}r^T S^{-1} r\right) \exp\left(-\frac{1}{2}u^T H^{-1} u\right)$$

coefficient — The coefficient is still the joint-density coefficient card

$$C = (2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}}$$

kp — Key points key-points

The two exponentials have the required shapes. To make them two complete Gaussian densities, C must supply the coefficient belonging to each dimension and covariance.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall, kp	—	Initial meaningful context
STATE 2	+ product, coefficient	entry emphasis	Apply the elementary exponential identity while visibly retaining C.

The same elimination gives the determinant product

Teaching move

Use the same block transformation to calculate the determinant appearing in the remaining coefficient.

Layout sketch

recall

triangular

product

kp

Content objects

recall — Carried forward recall-card

$$D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}, \quad T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}$$

From: P4 and P7

triangular — T is block triangular; determinants multiply card

$$\det T = \det I_{d_1} \det I_{d_2} = 1$$

$$\det D = \det T \det \Sigma \det T^T = \det \Sigma$$

product — D is block diagonal card

$$\det D = \det S \det H \quad \Rightarrow \quad \det \Sigma = \det S \det H$$

kp — Key points key-points

The coefficient contains $(\det \Sigma)^{-\frac{1}{2}}$. This identity replaces it by the determinant factors for the residual and observed blocks.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall, kp	—	Initial meaningful context
STATE 2	+ triangular	entry emphasis	Unit determinant means the elimination does not change det Sigma.
STATE 3	+ product	entry emphasis	Read the determinant of the block diagonal result.

Design constraints

All determinants are positive by the already proved strict positivity. This is the coefficient calculation promised on P2/P11, not an isolated matrix lemma.

The factors before the exponentials split too

Teaching move

Account for both determinant and dimension factors in the full coefficient.

Layout sketch

recall

split

coefficients

kp

Content objects

recall — Carried forward recall-card

$$d = d_1 + d_2, \quad \det \Sigma = \det S \det H, \quad S, H \succ 0$$

From: P1, P8 and P12

split — Add the dimensions; use positive determinant square roots card

$$(2\pi)^{-\frac{d}{2}} = (2\pi)^{-\frac{d_1}{2}} (2\pi)^{-\frac{d_2}{2}}$$

$$(\det \Sigma)^{-\frac{1}{2}} = (\det S)^{-\frac{1}{2}} (\det H)^{-\frac{1}{2}}$$

coefficients — Group each covariance with its matching dimension card

$$C = C_1 C_2$$

$$C_1 := (2\pi)^{-\frac{d_1}{2}} (\det S)^{-\frac{1}{2}}$$

$$C_2 := (2\pi)^{-\frac{d_2}{2}} (\det H)^{-\frac{1}{2}}$$

kp — Key points key-points

The residual exponential receives C_1, the coefficient for dimension d_1 and covariance S. The observed exponential receives C_2, for dimension d_2 and covariance H.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall, kp	—	Initial meaningful context
STATE 2	+ split	entry emphasis	Expose both pieces of the coefficient; positivity justifies the square-root identity.
STATE 3	+ coefficients	entry emphasis	Pair dimension and covariance before reassembling densities.

Reassemble the complete Gaussian densities

Teaching move

Pair each coefficient with its exponential and establish exact normalized factorization before naming probability laws.

Layout sketch

DESIGN LAYOUT

Adjacent Gaussian-factor regions, each with its own coefficient/exponent pairing, on a common baseline. First factor in x1 on the left; observed-block factor on the right. Product equality below both. No extra bordered wrapper around every formula. Keep lower interpretation separate and keep its prose in normal paragraphs. C1 and C2 may be expanded instead of abbreviated if readable; their definitions from P13 must remain available.

Content objects

factor1 — Gaussian density in x1 for fixed x2 — card

$$g_1(x_1; x_2) := C_1 \exp\left(-\frac{1}{2} r^T S^{-1} r\right)$$
$$= \mathcal{N}_{d_1}(x_1; m(x_2), S)$$
$$m(x_2) = \mu_1 + B(x_2 - \mu_2), \quad r = x_1 - m(x_2)$$

factor2 — Gaussian density in the observed block — card

$$g_2(x_2) := C_2 \exp\left(-\frac{1}{2} u^T H^{-1} u\right)$$
$$= \mathcal{N}_{d_2}(x_2; \mu_2, H), \quad u = x_2 - \mu_2$$

product — The exponential and coefficient products both match — card

$$p(x_1, x_2) = g_1(x_1; x_2) g_2(x_2)$$

kp — Key points — key-points

These are complete Gaussian densities because their coefficients and positive definite covariances match their dimensions. Integration and division will identify which factor is the marginal and which is the conditional.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	factor1, kp	—	Initial meaningful context
STATE 2	+ factor2	entry emphasis	Match the observed-block coefficient and exponential.
STATE 3	+ product	entry emphasis	Combine P11 exponential equality with P13 coefficient equality.

Design constraints

No proportionality sign. Retain x2 dependence in g1. The two factors are not yet named conditional/marginal. If C1/C2 are expanded, retain matching d1,S versus d2,H.

Integrate over x1 to identify the marginal

Teaching move

Use Gaussian normalization in the missing block to identify g2 as the observed marginal.

Layout sketch

recall

normalization

marginal

kp

Content objects

recall — Carried forward recall-card

$$p(x_1, x_2) = g_1(x_1; x_2) g_2(x_2)$$

From: P14

normalization — Integrate the Gaussian density in x1 card

$$x_2 \text{ fixed}, \quad m(x_2) \text{ fixed}, \quad S \succ 0$$

$$\int_{\mathbb{R}^{d_1}} g_1(x_1; x_2)\, \mathrm{d}x_1 = 1$$

marginal — g2 is constant with respect to x1 card

$$p(x_2) = \int_{\mathbb{R}^{d_1}} p(x_1, x_2)\, \mathrm{d}x_1$$

$$= g_2(x_2) \int_{\mathbb{R}^{d_1}} g_1(x_1; x_2)\, \mathrm{d}x_1 = g_2(x_2)$$

$$p(x_2) = \mathcal{N}_{d_2}(x_2; \mu_2, H)$$

kp — Key points key-points

For each fixed observation, the first factor integrates to one. Marginalizing the joint therefore leaves exactly the second factor.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall, normalization, kp	—	Initial meaningful context
STATE 2	+ marginal	entry emphasis	Apply the marginal-density definition and move the x1-constant factor outside the integral.

Design constraints

Keep the domain, integration variable and fixed observation visible. Normalization of a Gaussian density is prior knowledge; no Jacobian proof is needed.

Divide by the marginal to obtain the conditional density

Teaching move

Use the now-identified strictly positive marginal to obtain the conditional-density formula.

Layout sketch

positive

divide

law

kp

Content objects

positive — Carried forward recall-card

$$p(x_2) = g_2(x_2) = \mathcal{N}_{d_2}(x_2; \mu_2, H) > 0$$

From: P15; H strictly positive definite

divide — Use the conditional-density ratio; cancel the positive factor card

$$p(x_1|x_2) = \frac{p(x_1, x_2)}{p(x_2)}$$

$$= \frac{g_1(x_1; x_2) g_2(x_2)}{g_2(x_2)} = g_1(x_1; x_2)$$

law — The Gaussian conditional density card

$$p(x_1|x_2) = \mathcal{N}_{d_1}(x_1; \mu_1 + B(x_2 - \mu_2), S)$$

kp — Key points key-points

The factor identified by division is already normalized in x_1. This establishes the complete conditional density for every observed value x_2.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	positive, kp	—	Initial meaningful context
STATE 2	+ divide	entry emphasis	Identify the first factor from the joint and the independently established marginal.
STATE 3	+ law	entry emphasis	Write the resulting law with the proved parameters.

Design constraints

Positivity is a density statement, not positive probability of a continuous singleton. This defines a conditional-density version everywhere; do not assert uniqueness at null events.

The observation can shift the conditional mean

Teaching move

Interpret where observation dependence resides in the proved law, distinguishing original blocks from residual coordinates.

Layout sketch

law

dependence

kp

Content objects

law — Carried forward recall-card

$$p(x_1|x_2) = \mathcal{N}_{d_1}(x_1; \mu_1 + B(x_2 - \mu_2), S)$$

From: P16

dependence — A changed observation shifts the mean; the covariance stays S card

$$m(x_2) = \mu_1 + B(x_2 - \mu_2)$$
$$m(x_2 + \Delta) - m(x_2) = B\Delta$$
$$\Sigma_{1|2} = S = \Sigma_{11} - \Sigma_{12}H^{-1}\Sigma_{21}$$

kp — Key points key-points

The mean can change with x_2 through B; S does not depend on the observed value.
The residual $r = x_1 - m(x_2)$ uses the observation. Writing its quadratic separately does not make the original X_1 and X_2 independent.
The result is exact for the supplied Gaussian model with $\Sigma \succ 0$; it does not test that model against data.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	law, kp	—	Initial meaningful context
STATE 2	+ dependence	entry emphasis	Vary the observation in the proved mean formula while holding model parameters fixed.

Design constraints

Final slide is interpretation, not a disguised section summary. “Can change” includes B=0 and directions in its kernel. Delta has dimension d2. No new algorithm, singular extension, notebook handoff, or rehearsal claim.